9(a)(i)	Draw a smooth curve/best fit line.	[1]
(ii)	Draw a horizontal line where the curve approaches the asymptote to obtain the background count rate.	[1]
(iii)	From the graph, background CR ~ 10.0 min^{-1}. CR on graph = 170 min^{-1} , (Corrected) CR due to source only = 170 – 10 = 160 min^{-1}. After $t_{1/2}$, (Corrected) CR due to source only = 160/2 = 80 min^{-1}, CR on graph = 80+10 = 90 min^{-1} , **For CR on graph to reduce from 170 to 90 min^{-1} ,** $t_{1/2}$ ~ 1.6 hours Need to find the half-life using another set of points from the graph, and then find the **average** half-life: CR on graph = 120 min^{-1} , (Corrected) CR due to source only = 120 – 10 = 110 min^{-1}. After $t_{1/2}$, (Corrected) CR due to source only = 110/2 = 55 min^{-1}, CR on graph = 55 + 10 = 65 min^{-1} , **For CR on graph to reduce from 120 to 65 min^{-1} ,** $t_{1/2}$ ~ 1.3 hours Hence averaged $t_{1/2}$ ~ 1.45 hours	[1] [1] [1]
(iv)	half-life: no change, because decay is spontaneous / is independent of environment / external factors like temperature.	[0] or [2]

(b)(i)	Total momentum before decay = total momentum after decay 0 = 4 (+V) + (A – 4](-v) -------- eqn (1) Marking: no mark if negative sign is omitted	[1]
(ii)	LHS = $\frac{\frac{1}{2}m_\alpha V^2}{\frac{1}{2}m_D v^2} = \frac{4V^2}{(A-4)v^2}$ -------- eqn (2), m_D= mass of daughter nucleus From eqn (1) in (b)(i), $\frac{V}{v} = \frac{A-4}{4}$ Sub eqn (1) into eqn (2), LHS = …. = ($\frac{1}{4}$A – 1) (Shown) or KE = $\frac{p^2}{2m}$ {Allowed to recall & use} Thus $\frac{KE_{\alpha,i}}{KE_{D,i}} = \frac{p_\alpha^2}{2m_\alpha} \div \frac{p_D^2}{2m_D}$ where $p_\alpha = p_D$ {by PCLM} $= \frac{m_D}{m_\alpha} = \frac{(A-4)}{4} = \frac{A}{4} - 1$ (Shown)	[1] [1] [1]
(c)(i)	$^{212}_{83}Bi \rightarrow {}^{208}_{81}Tl + {}^{4}_{2}\alpha$ Energy released = $\Delta m\, c^2$= { 211.9459u – [207.9374*u* + 4.0015*u*] } c^2 = 0.007uc^2 {*u*: value taken from List of Data} = 1.0458 x 10^{-12} J = 6.54 MeV 1 MeV = 1.6 x 10^{-13} J	[1] [1] [1] [1]
(ii)	{Use (c)(i) & (b)(ii) to show energy of α-particle is 6.42 MeV} Let initial KE of α = x. Thus initial KE of daughter nucleus (i.e. Tl) = 6.54 – x MeV From (b)(ii), $\frac{x}{6.54-x} = \frac{212}{4} - 1$ Solving, x = 6.42 MeV (shown)	[1] [1]

(d)(i)	Some of the decay energy is carried away as gamma radiation, so the alpha-particle has less kinetic energy.	[1]
(ii)	After decay, the total momentum of the α-particle, thallium nucleus & γ = 0 by conservation of linear momentum. For thallium nucleus & the α-particle to move off in opposite directions, all 3 particles will have to move off along the same line by the principle of conservation of momentum, which is not likely. Concept: Note that the γ -photon will therefore likely have a component of momentum in the direction perpendicular to the velocities indicated in the qn. PCLM must also hold for this perpendicular direction as well. The thallium nucleus and α-particle therefore are unlikely to move off in opposite directions.	[1] [1]